Data Mining

Predicting Used Car Prices on Craigslist: An iterative Machine Learning Approach, Github: https://github.com/sinandraide/Used-Car-Data-Mining



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1 Introduction

1.1 Problem and Motivation

The used car market represents a dynamic and complex segment of the automotive industry, where pricing decisions are influenced by a multitude of factors such as vehicle condition, mileage, brand, model year, fuel type, and location. For both buyers and sellers, accurately estimating a fair market value is a crucial yet challenging task. Unlike new vehicles, where prices are more standardized, used car prices vary widely and often lack transparency, leading to inefficiencies and suboptimal transactions.

This project focuses on addressing the challenge of predicting the price of used cars using the benchmark "Used Car Dataset" from Kaggle which is based on raw observations from Craigslist — one of the largest online second-hand shopping platforms for vehicle listings in the United States. The data includes a diverse and unstructured collection of real-world listings, encompassing numerous makes, models, and regions, along with noisy, missing, or inconsistent values typical of user-generated content. These characteristics introduce significant challenges in data preprocessing, feature selection, and model generalization.

Beyond building a predictive model, the project also aims to identify which vehicle attributes most strongly influence price. This has practical implications not only for consumers seeking fair prices but also for dealerships, platforms, and data-driven pricing engines aiming to provide accurate recommendations.

By leveraging machine learning and data mining techniques, this project aims to bring structure and insight to the unstructured nature of Craigslist data, providing value to multiple stakeholders in the used car market.

1.2 Objectives

The main objectives of this project are:

- **Address Data Challenges**: Handle real-world data issues including missing values, inconsistent entries, categorical encoding, and data imbalance.
- **Develop a Predictive Model**: Train machine learning models capable of accurately predicting the price of used vehicles listed on Craigslist based on their features.
- **Identify Key Price Drivers**:Analyze feature importance to determine which attributes — such as mileage, manufacturer, condition, or model year — have the greatest impact on vehicle pricing.
- **Iteratively Improve Performance**: Use an iterative approach to refine data preprocessing, model selection, hyperparameter tuning, and evaluation in order to enhance the model's predictive accuracy.

- **Evaluate and compare model Performances**: Compare different modeling approaches (e.g., linear regression, tree-based models, ensemble methods) and validate their performance using appropriate metrics such as MAE, RMSE, and R^2.
- **Evaluate Model Robustness:** Following the training and evaluation phases, we assess how well the models handle anomalies and outliers to determine their resilience and generalizability.

2 Data Profiling

2.1 Structure and Size of the Dataset

Before any individual interested in selling a car can place a listing on Craigslist, the potential seller can include multiple attributes pertaining to the vehicle itself and its condition such as manufacturer, model, condition, and the odometer values. Table 2.1 provides further insights into key attributes which may be utilized as features for model training. The table exlcudes meta data including unique identifiers, county, longitude and latitude of the listed vehicle. The dataset contains roughly 426,000 listings of used vehicles and 26 attributes, spanning from Hawaii and Alaksa to the eastern coast of the United States with the most common region, with a frequency of 3608 cars being listed in the Columbus Metropolitan Area of Ohio.
At first glance, the raw dataset shows signs of poor quality, with several indicators of extreme outliers, missing data, and potential input errors that warrant closer inspection. For example, the target attribute price displays a highly skewed distribution. Prices range from 0 to a staggering 3.736 billion dollars, with a mean of about 75,199 and a median of just 13,950 dollars. This stark contrast between the mean and median reflects the impact of extreme outliers. The standard deviation exceeds 12 million dollars, and the skewness is exceptionally high at 254.4, indicating that a small number of listings are distorting the distribution significantly. In total, 196 records have an asking price of over 150,000 dollars which could account for noise or unrealistic offers. On the other hand, 7844 listings have a price of zero while not including information on manufacturer, model, or year which again may highlight data quality issues.
Similarly, the odometer field which represents vehicle mileage ranges from 0 to 10 million miles. The average mileage is 98,043, while the median is 85,548, and the standard deviation is over 213,000 miles. Like price, this field is also heavily right-skewed, with a skewness of 38.0, again indicating the presence of extremely high or possibly erroneous values. Other key attributes include year, which ranges from 1900 to 2022, with a mean of 2011, and manufacturer, where Ford appears most frequently. The dataset includes nearly 30,000 unique models, led by the "F-150", and fuel type is predominantly gas. Additionally, an overwhelming majority of 94169 listed cars have six cylinders. Among vehicles with color information, white is the most common.
Another problem in the dataset is the amount of missing values for some attributes. Attributes with high missing values are for example: county (100 percent), size (71.77 percent), cylinders (41.62 percent), condition (40.79 percent), VIN (37.73 percent), drive

(30.59 percent), paint color (30.5 percent), and type (21.75 percent). But there are also attributes with low missing values like fuel (0,01 percent), year (0.28 percent), model (1.24 percent) or manufacturer (4.13 percent).
The dataset contains the following key attributes:

Attribute	**Type**	**Description**
`price`	Numerical	Listed price of the vehicle (target variable)
`year`	Numerical	Year the vehicle was manufactured
`manufacturer`	Categorical	Brand of the vehicle (e.g., Ford, Toyota)
`model`	Categorical	Specific model name (e.g., Civic, F-150)
`condition`	Categorical	Vehicle condition (e.g., new, like new, fair)
`cylinders`	Categorical	Engine configuration (e.g., 4 cylinders, 6 cylinders)
`fuel`	Categorical	Fuel type (e.g., gas, diesel, electric)
`odometer`	Numerical	Miles the vehicle has been driven
`title_status`	Categorical	Legal status of the vehicle title (e.g., clean, salvage)
`transmission`	Categorical	Type of transmission (e.g., automatic, manual)
`drive`	Categorical	Drivetrain (e.g., 4wd, fwd, rwd)
`type`	Categorical	Vehicle type (e.g., SUV, truck, sedan)
`paint_color`	Categorical	Exterior color of the vehicle
`state`	Categorical	U.S. state where the listing was posted
`region`	Categorical	Specific geographical area or city within a state
`VIN`	Text / Identifier	Unique vehicle identification number; can be used to retrieve external vehicle history data
`posting_date`	Date	Date of the Craigslist posting

Table 1: Overview of Key Attributes in the Craigslist Cars and Trucks Dataset

2.2 Attribute Relevance and Domain Knowledge

Attribute	Expected Impact on Price	Domain Reasoning
`year`	Positive	Newer vehicles are typically more valuable due to lower wear and newer features.
`odometer`	Negative	Higher mileage usually indicates more usage and wear, reducing the car's value.
`manufacturer / model`	Mixed	Brand reputation and demand affect pricing; premium brands typically retain value better.
`condition`	Positive	Better condition implies less damage and wear, thus higher price.
`fuel`	Mixed	Some fuel types (e.g., electric, hybrid) may increase price due to efficiency and market trends.
`transmission`	Slight Positive	Automatic transmissions are often preferred, especially in the U.S.
`title_status`	Negative (if salvage)	Vehicles with a salvage or rebuilt title generally have a significantly lower market value.
`drive`	Context-dependent	4WD/AWD systems may add value in regions with harsh weather or for certain vehicle types.
`type`	Mixed	Trucks and SUVs may retain more value due to higher utility, depending on region.
`region / state`	Context-dependent	Urban or high-income areas may see higher prices; vehicle preferences also vary by region.
`VIN`	Indirect	VIN is not a price driver itself but can be used to access service records and accidents, which may influence perceived value.

Table 2: Expected Impact of Key Attributes on Vehicle Price Based on Domain Knowledge

As stated in the project objectives, this study aims to identify the key variables that influence the target attribute, price. Table 2.2 provides reasons explaining how certain attributes may affect price. It is expected that a higher manufacturing year, better vehicle condition, and having an automatic transmission will have a positive effect on price. In contrast, higher odometer readings and some categories within the "title_status" attribute, such as "salvage" are likely to have a negative effect. The opposite effects are expected when these variables decrease.

Among all the attributes, we hypothesize that year, odometer, and condition will have the strongest impact on price. This is supported by the heatmap correlation matrix shown in Figure 1, where odometer and year have the highest positive and negative correlation coefficients: odometer shows a moderate negative correlation (-0.46) with price, while year shows a weak positive correlation (0.32). Surprisingly, the vehicle's

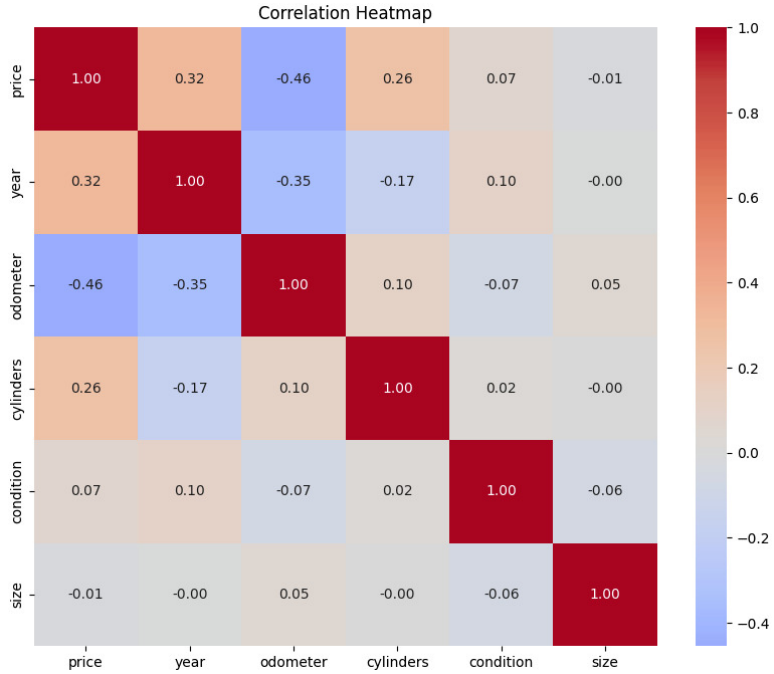


Figure 1: Correlation Heatmap

"condition" does not appear to correlate with price.
These results are somewhat consistent with those of Fatima et al. (2024, pp. 161–163), who studied the effects of similar attributes on used car prices. Their analysis found a stronger negative correlation between odometer and price (-0.66) compared to our results. This difference may be due to the quality of the dataset. Overall, while the dataset is rich and detailed, it contains substantial outliers and skewness in major numeric fields, potential noise points and many missing values, making data cleaning a necessary step before training any models.

3 Preprocessing

Our preprocessing strategy addresses multiple data quality challenges in the Craigslist vehicle dataset to create a reliable foundation for our modeling. We implement methodical approaches for handling missing values, outliers, and inconsistencies across attributes, significantly improving data quality while maintaining dataset integrity. Before applying these strategies however, we use the hold-out method to divide the dataset into training and testing sets with an 80/20 split, avoiding any data leakage that may occur while preprocessing and ensuring the test set remains unseen.

3.1 Handling Missing Values

We developed attribute-specific imputation strategies based on data patterns and domain knowledge:
For the drive attribute, we observe strong correlations between vehicle types and drive systems (i.e. minivans: 92% fwd; offroad: 92% 4wd; coupe: 70% rwd). Given these patterns, we impute missing values using the mode per vehicle type. For the manufacturer and model attributes, we implement a multi-step approach using VIN numbers (first digits identify manufacturers, digits 4–8 for models), description text mining, and consistent pattern mapping. This resolves approximately 12,800 missing manufacturer values and 1,400 missing model values. For attributes such as title status and condition, we avoid potentially biased assumptions by marking missing values as "missing" rather than defaulting to dominant categories like "clean" or "good" condition, preserving data integrity. To handle missing values in size and type, we develop hierarchical approaches combining model-level patterns, manufacturer tendencies, and logical mapping rules (i.e. SUV: full-size, sedan: mid-size). For critical pricing factors like fuel type and odometer, we removed rows with missing values (7,400 total) rather than risk introducing any biases through imputation.

3.2 Outlier Detection and Handling

As discussed before, the target attribute, price, initially shows extreme right skewness in the raw dataset. In the training set however, the skewness increases to 333.54, given a mean of 51,920, a median of 13,995, and standard deviation of 9,216,972 dollars, explaining the high error in initial models in the later sections. After testing various approaches, we apply a 3-IQR method (factor of 3 instead of 1.5) to identify and remove 669 price outliers beyond -55,790 to 88,375 dollars. This conservative modification of the IQR on the one hand removes statistical anomalies while preserving legitimate higher-priced vehicles. Post outlier removal, the skewness shrinks to 1.10 and the standard deviation to 14,524 dollars.
For odometer readings, which include implausible values up to 10 million miles, we apply the standard 1.5-IQR rule to remove approximately 3,100 entries. This normalizes the distribution (skewness reduced from 40.53 to 0.48), with a mean of 89,931 miles and median of 85,000.

3.3 Data Validity and Feature Engineering

We implement logical consistency checks to identify problematic listings, such as new vehicles with excessive mileage or discrepancies between condition and price. These checks identify approximately 2,500 questionable listings that are removed. Duplicate detection identifies approximately 4,200 listings through exact and near-duplicate matching, which are removed to prevent data leakage between training and testing sets.
Before training a model, we apply a final round of preprocessing steps to ensure that the input can be used for the appropriate model. As such, we set up a pipeline in which we encode categorical and scale numeric attributes. Next, we select the models and

hyper-tune the parameters using a random search method. For each hyper-tuned model, we then perform a permutation importance analysis to determine which attribute the model learnt the most from. Finally, we conduct a statistical comparison between a median price baseline model and the trained models.
For the encoding part, we use an encoding method based on the properties of each attribute, such as its cardinality and whether it possesses an inherent order. We use a binary encoder for the "region" attribute because it has high cardinality (400 unique regions) and is nominal, as the regions have no internal order. Due to the high cardinality, one-hot encoding would not be effective. For the attributes "manufacturer"", "state"" and "model", the chosen encoding method is one-hot encoding. Like ""region"", these attributes have a small number of unique categories (a maximum of 13). They also have no internal order.
The final attributes, "condition" and "size"", are ordinal encoded because they have an inherent order. Examples of 'condition' values include "fair", "good" and "excellent", and examples of "size" values include "compact", "mid-size" and 'full-size'. For the numerical values ""year"" and "odometer", we use the standard scaler method to normalize the data.

4 Modeling - An Iterative Journey

For the hyperparameter tuning, we use random search since it offers a good balance between performance and results. We evaluate the models using cross-validation. It is also computationally less expensive than nested cross-validation. We use 50 iterations for random search and 5-fold cross-validation to achieve a good balance between performance and results.
In order to evaluate the performance, we use common evaluation metrics for regression: Mean Absolute Error (MAE), Coefficient of Determination (R^2), Root Mean Squared Error (RMSE) and the Median Absolute Error (MDE) and Mean Squared Error (MSE). We run three tests on the following pre-selected models: linear regression, polynomial regression, regression tree, random forest regression and Xtreme Boosting. We select these models because each has its own strengths and weakness regarding linear and non-linear problems. Additionally, they do not require intensive computing power. After the first run, in which the models were trained on the pre-processed training set which contained extreme outliers for price and odometer, we obtained enormous values for the chosen evaluation metrics; for example, the MSE for linear regression was 281.4 trillion.

Model	Test MAE	Test MedAE	Test MSE	Test RMSE	Test R2 Score	Evaluation Time (s)
linear_simple	166117.364	60853.48224	281496468487380	16777857.03	-3.76E-05	0.2940783501
linear_poly	331765.4806	172336.7818	281632992889558	16781924.59	-0.0005225437856	1.827973366
dt	126377.27	9988.529803	282033671514265	16793858.15	-0.001945984955	0.3193211555
rf	124479.2464	11835.62189	281499431708006	16777944.8	-4.81E-05	0.3240456581
xgb	120827.4766	24173.60547	281502088691712	16778023.98	-5.75E-05	0.3261702061

Table 3: Endresults for Literation 3

MSE calculates the average of the squared differences between the actual and predicted values. The logical consequence is that in the training data, the target value must have extreme outliers. As such, we remove any price that falls outside of the modified IQR for the next iteration as discussed in the preprocessing section.
However, when looking at the permutation importance, we discovered that the odometer variable had huge negative values.

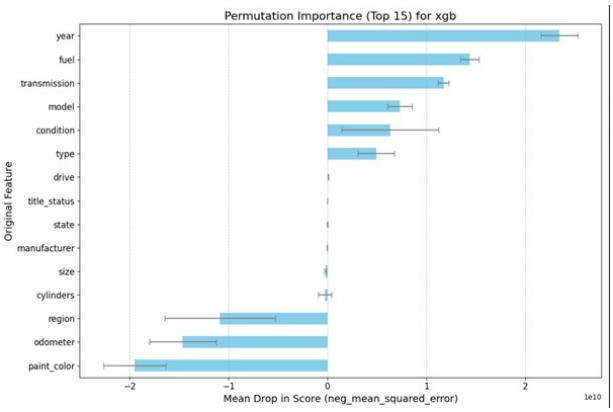


Figure 2: Permutation Importance

In the section iteration, when looking at the permutation importance, we discover that the odometer variable, too, contains huge negative values. As a result, we also remove extreme outliers which fall outside of the 1.5-IQR range. After having removed said outliers from price and odometer, we re-train the models again. Now, we have much better results: the MSE for the linear regression decreased to 114.2 million. However, the metrics are still not good.
In the third iteration, we apply a log transformation on the dependent variable, price, and thus restrict the dataset to entries whose price are greater than zero to avoid undefined values during the transformation. This reduces the effect of extreme price values and distributions that are only slightly right skewed. Log-transforming the target variable helps regression models to better capture relative differences rather than absolute differences, particularly when price values vary across several orders of magnitude. It also leads to more normally distributed residuals. After training the model on log-transformed prices, we use an exponential function to convert the predicted values back to their original scale. This post-processing step enables us to interpret the results in line with the original pricing domain. The table below shows the results from this third run. The model now outputs more realistic-looking predictions, especially for higher-priced cars.
The table above shows the results from this third run. The model now outputs more realistic-looking predictions, especially for higher-priced cars.
To compare the performance of our models against a baseline, we apply two statistical tests: the paired t-test and the Wilcoxon signed-rank test. These tests are used to determine whether the differences in prediction errors between each model and the baseline

Model	Train Log R^2	Train Log RMSE	Train Log MSE	CV Log RMSE	CV Log MSE	Test R^2	Test RMSE	Test MSE	Test MAE	Test MedAE	Eval Time (s)
linear_simple	0.2899	1.0322	1.0654	1.0325	1.0654	0.5238	9811.5146	96265818296411 10.0000	6621.2400	4163.4400	0.2592
linear_poly	0.3713	0.9713	0.9433	0.9714	0.9435	0.5239	9810.4468	96244866851259 00.0000	6231.4200	3803.9600	1.3516
dt	0.7436	0.6202	0.3846	0.8310	0.6906	0.7346	7324.9253	53654530529811 40.0000	3834.2900	1720.8700	0.3044
rf	0.4463	0.9114	0.8307	0.9227	0.8514	0.5660	9065.3590	87728806747337 00.0000	5970.5800	3422.0300	0.3072
xgb	0.5914	0.7830	0.6130	0.8022	0.6435	0.7002	7784.8644	60604114205149 00.0000	4736.2700	2647.9300	0.3261
baseline (median)	–	–	–	–	–	-0.0507	14573.2532	21237970997639.0000	11139.1813	9400.0000	0.3265

Table 4: Updated Evaluation Metrics Based on Screenshot

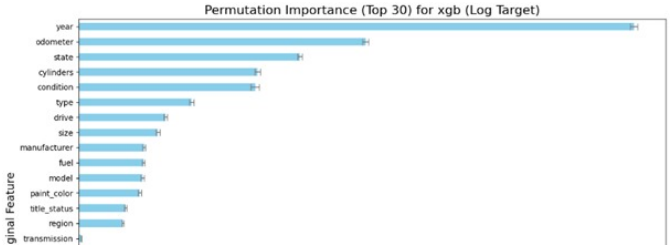


Figure 3: Permutation Importance

are statistically significant.
As a baseline model, as discussed before, we use the median car price of 15,590 dollars. This represents a naive predictor that always returns the same fixed value.

Comparison	T-statistic	T P-value	Wilcoxon-statistic	Wilcoxon P-value	T-test Significant	Wilcoxon Significant
Median Price Baseline vs linear_simple	161.22	0	518050177.00	0	Yes	Yes
Median Price Baseline vs linear_poly	162.98	0	472266728.00	0	Yes	Yes
Median Price Baseline vs dt	206.16	0	330262724.00	0	Yes	Yes
Median Price Baseline vs rf	206.16	0	342933904.50	0	Yes	Yes
Median Price Baseline vs xgb	223.08	0	303816351.50	0	Yes	Yes

Table 5: Statistical Comparison: Median Price Baseline vs Models

The table above presents the results. As shown, both the t-test and Wilcoxon test indicate that all trained models significantly outperform the baseline. The p-values for all comparisons are effectively zero. This strongly suggests that the differences in performance are not due to random chance.
For the decision tree, we generate a symbolic tree with a maximum depth of four. The decision tree begins by evaluating the 'num_year' feature at a depth of 4 and with a threshold of 0.12. This initial split categories the data based on this criterion. Following this, the tree conducts further splits. On the left branch of the initial split, 'num_year' is evaluated again with a threshold of 0.22. The data is then divided based on the 'drive' feature using the 'onehot_enc_drive_fwd' dummy variable. On the right branch, the tree examines additional numeric features. It considers 'num_odometer' and 'num_cylinders' to refine predictions. Each of these conditions leads to two branches: one for cases where the condition is true, and one for cases where it is false. They also show that the 'odometer' and 'year' variables have a significant impact on model prediction, as described by the permutation importance.

5 Results

Out all models, the decision tree has the best performance. It has the highest R^2 score and explains approximately 73.5% of the variance given unseen data. While this falls short of the performance reported by Fatima et al. (2024, p. 164), whose model achieved an R^2 of 0.86, the difference is relatively modest, approximately 13 percentage points. This suggests that our model performs competitively and is approaching the state of the art. Furthermore, the decision tree has the lowest errors on the original price scale across all primary error metrics (test RMSE: 7,324.9; test MAE: 3,834.29; test MedAE: 1,720.87), indicating that it has the lowest error.

XGBoost has the second highest test R^2 score (0.7002) and the second lowest error metrics (test RMSE 7,784.86, test MAE 4,736.27 and test MedAE 2,647.93). The baseline model has a Test R^2 score of -0.0507; the negative R^2 indicates that the baseline model performs worse than simply predicting the mean of the test set values. In fact, all models outperform the baseline, which shows that all models are learning meaningful patterns.

For all models, there is a gap between test and training scores, especially when the scales change, as with log transformation. However, for the Decision Tree, the train log RMSE (0.6202) is considerably lower than the cross-validation log RMSE (0.8310). This clearly indicates that the decision tree overfit the log-transformed training data. The Xtreme Train Log RMSE (0.7830) is fairly close to the Cross Validation Log RMSE (0.8022). This suggests that Xtreme is less overfitted to the log-transformed training data than the decision tree. Random forests did not overfit much and linear regression models almost not at all.

Linear models perform much better than the baseline model, but not as well as the tree-based models. Polynomial regression offers a slight improvement versus the simple linear model. This suggests that non-linear relationships are important in our data and that the tree-based models capture these better. However, the permutation importance results highlight 'year' and 'odometer' as the top raw features which goes in line with the correlation analysis explained earlier. Nevertheless, it is important to recognise the collective impact of the categorical features. These were transformed into numerous dummy variables via one-hot encoding (and other strategies, such as binary encoding for 'region'). While individual dummy variables might not rank as highly as 'year' or 'odometer' in permutation importance, their combined presence enables the models, particularly the tree-based ones, to capture specific nuances and segmentations within the data. This contributes significantly to the models' strong overall fit and predictive power. Tree-based models are adept at utilising these many binary features to define complex decision boundaries and interactions.

It is noticeable that the random forests perform worse than the decision tree, which is unusual for an ensemble model. This suggests that the hyperparameter tuning parameters might have been poorly chosen for the random forest. Despite the lower scores campared to the decision tree , XGBoost was the final model we chose because it overfits less.

We also implemented a robust test to generate absurd values for each attribute and see how our models predict them. We used numerically extreme values, such as a very high

or very low odometer reading or year (e.g. year 1900 or 2077, or an odometer reading of 1 mile or 5 million miles). For categorical attributes, we used values such as a 'new' condition for a 1950s car or a 'truck' with '2 cylinders'. We also included missing values in many places. We also used values far outside the training distribution.
After running the tests, we discovered that the models cannot handle NaN. The ordinal encoder 'condition' and 'size' cannot handle NaN. For all examples, the ordinal encoder will match if it has a class that has been seen before. The same applies to binary encoding and frequency encoding. Further work is required for model deployment, and the encoding pipeline must be configured to handle this situation.
We ran the robots again with the following values:

Target_Price	Description	region	year	manufacturer	model	condition	cylinders	fuel	odometer	title_status	transmission	drive	size	type	paint_color	state
0 5000	New Truck, High Mileage	california	2025.0	ford	f-150	new	4	diesel	250000.0	clean	manual	rwd	mid-size	truck	yellow	ca
1 45000	Classic Car, 'New', 1 Mile	new york	1965.0	chevrolet	impala	new	8	gas	1.0	clean	automatic	rwd	full-size	convertible	black	ny
2 7000	Recent Salvage, 'Excellent', Low Miles	texas	2023.0	honda	civic	excellent	4	gas	500.0	salvage	automatic	fwd	compact	sedan	silver	tx
3 3000	Prius as V10 Diesel Pickup	california	2018.0	toyota	prius	good	10	diesel	60000.0	clean	manual	4wd	full-size	pickup	green	ca
4 10000	Porsche 911 as Basic FWD Van	florida	2020.0	porsche	911	fair	4	gas	75000.0	rebuilt	automatic	fwd	compact	van	white	fl

Table 6: Sample Vehicle Listings with Features and Target Prices

They describe different cars, but with random values that make no sense, in order to test the robustness of the model.
XGBoost has the lowest absolute price prediction error, followed by the decision tree, random forest and then the linear models, as described in the evaluation section. This robust test shows that the extreme boosting model is the best choice for predicting used car prices.

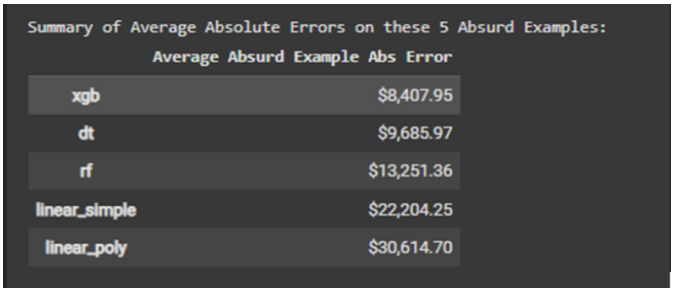


Figure 4: Summary of Absolute Errors

6 Summary

This project successfully transformed raw, noisy Craigslist data into a reliable basis for predicting used car prices. By implementing rigorous preprocessing—handling missing values, removing extreme outliers with modified IQR methods, and applying appropriate feature encoding—we effectively mitigated data inconsistencies and skewed distributions.

Our iterative approach demonstrated the limitations of simple linear models, while tree-based methods, particularly XGBoost, captured the important nonlinear relationships inherent in the data. Key features such as vehicle year, odometer readings, and various categorical attributes were confirmed as critical drivers of price. Despite some unexpected findings, like the lower-than-anticipated impact of vehicle condition, our final XGBoost model outperformed baseline predictions and other models in cross-validation and robustness tests. Future work can focus on improving model robustness against unseen values and integrating additional data sources or advanced NLP techniques to further refine predictive accuracy.

7 References / Sources

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8 Ehrenwörtliche Erklärung

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Declaration of Used AI Tools

Tool	Purpose	Where?	Useful?
ChatGPT	Rephrasing	Throughout	+
ChatGPT	Encoding assistance	Partially	+
DeepL	Translation	Throughout	+



Unterschrift
Mannheim, den 18.05.2025